Institut de Mathématiques de Toulouse, INSA Toulouse

Classification and Regression Trees

Machine Learning
4 MA - 2024-2025

Béatrice Laurent - Philippe Besse - Mélisande Albert - Cathy Maugis
Olivier Roustant

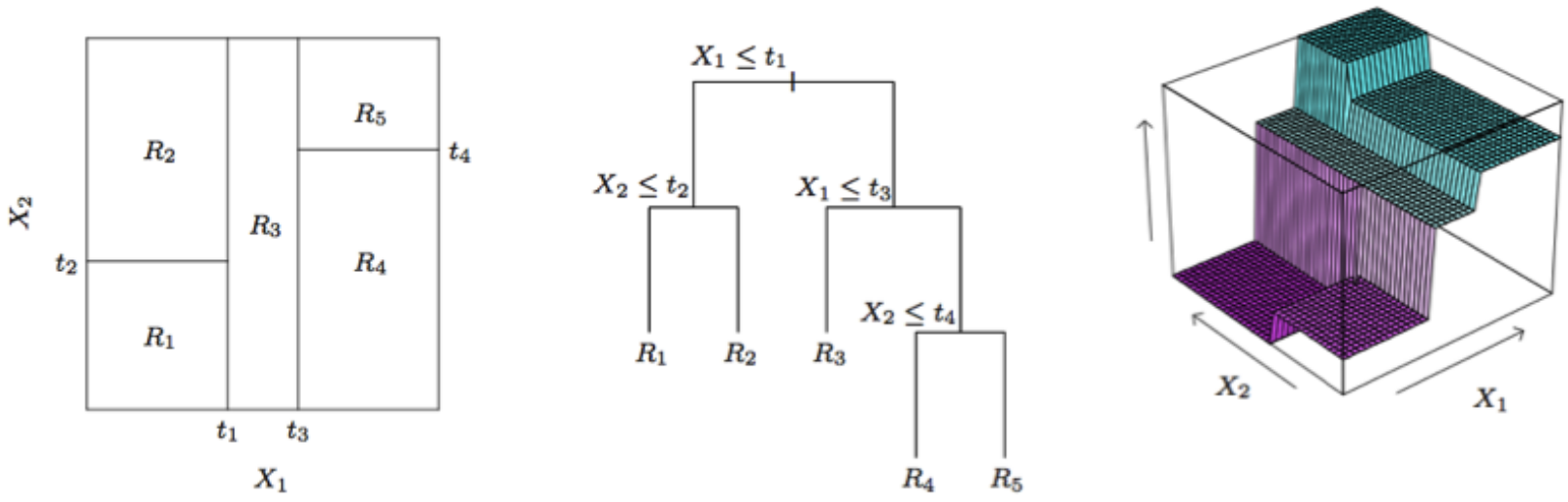
Outline

- Introduction
- Construction of a maximal binary tree
- Homogeneity criterion
- Pruning the maximal tree
- Practical remarks
- Conclusion

Introduction

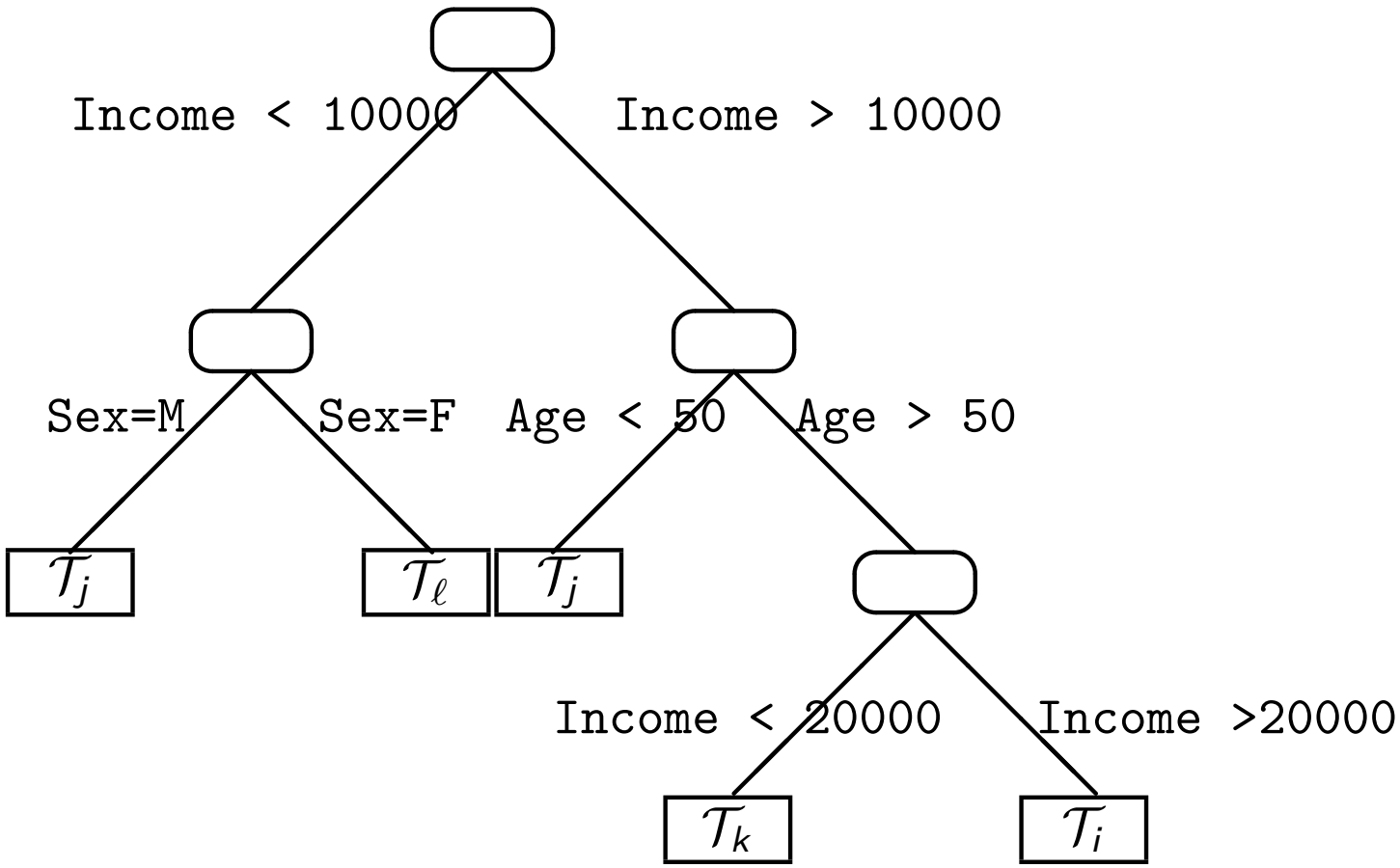
- The recursive partitioning or segmentation methods were first introduced in the 1960's.
- The method studied in this course was presented in a paper by Breiman et al. in 1984 under the acronym of CART for Classification and Regression Trees. This method can be used either for regression or for classification.
- The CART algorithm is a non parametric method to build estimators in a multidimensional framework.
- The method, based on trees, relies on a partition of the space of input variables. We then infer a simple model (constant piecewise functions in regression and a single class in classification) on each element of the partition.
- The obtained solutions can be represented in a graphic with a tree that is very easy to interpret. The trees are based on a recursive sequence of division rules or splits, each of them based on a single explanatory variable.

Example of binary regression tree



Source : Hastie, Tibshirani, Friedman (2019), "The elements of statistical learning"

Example of binary classification tree



Introduction

- A first very simple and natural non parametric procedure in supervised regression or classification is the k-Nearest Neighbors (k-NN) method.
- Given a leaning sample $\{(\boldsymbol{X}_1, Y_1), \ldots, (\boldsymbol{X}_n, Y_n)\}$ in $\mathcal{X} \times \mathcal{Y}$, we want to predict the output Y associated to a new entry $\boldsymbol{x}$.
- For this, it seems natural to built the predictor from the observations in the training sample that are "close" to $\boldsymbol{x}$.
- We consider a distance d on $\mathcal{X}$. We fix an integer k and we retain the k nearest to $\boldsymbol{x}$ observations $\{\boldsymbol{X}_{(1)}, \ldots, \boldsymbol{X}_{(k)}\}$ and the associated outputs $(Y_{(1)}, \ldots, Y_{(k)})$.
- In a regression context, the prediction at point $\boldsymbol{x}$ is obtained from the mean of the observations $(Y_{(1)}, \ldots, Y_{(k)})$ while in classification we consider a majority vote.
- The choice of k is of course crucial. A too small value leads to overfitting (small bias but high variance) while a large value of k may lead to underfitting (small variance but probably high bias).

Introduction

- CART uses the same idea of local mean or majority vote, but the cell in $\mathcal{X}$ that is used to predict at point $\boldsymbol{x}$ is obtained from a more sophisticated way than simply considering the k-Nearest Neighbors of $\boldsymbol{x}$ in the learning sample.
- It will also take into account the values of the Y_i's.
- When partitioning ends, each terminal node of the complete tree becomes a leaf to which is assigned a value if Y is quantitative and a class if Y is qualitative.
- The last step consists in pruning the complete tree, which corresponds to a model selection procedure in order to reduce the complexity and avoid overfitting.

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Principles for constructing a tree

- Recursive binary split
  $\rightarrow$ Split a region in two, then split subregions in two, then ...
- Splits are defined by one variable
  $\rightarrow$ Very easy numerically : p optimizations in 1-dimension
- Clustering idea
  $\rightarrow$ Find a split that give the most homogeneous groups

- We observe p quantitative or qualitative explanatory variables X^j and a variable to predict Y which is either qualitative with m modalities $\{\mathcal{T}_\ell; \ell = 1 \ldots, m\}$ or real quantitative, on a sample of n individuals.
- The construction of a binary discrimination tree consists in determining a sequence of nodes.
  - A node is defined by the choice of a variable among the p explanatory variables and of a division which induces a partition into two classes. Implicitly, to each node corresponds a subset of the initial sample to which a dichotomy is applied.
  - A division is defined by a threshold value if the selected variable is quantitative or a split into two groups of modalities if the variable is qualitative.
  - At the root, the initial node corresponds to the whole sample ; the procedure is then iterated over each of the subsets.

The algorithm requires :

1. the definition of a criterion allowing to select the best division among all admissible ones for the different variables
2. a rule allowing to decide that a node is terminal : it thus becomes a leaf
3. the predicted value (class or real value) associated to a leaf.

Division criterion

- A division is said to be admissible if the two corresponding son nodes are not empty.
- If the explanatory variable is a quantitative variable with m possible values (or qualitative but ordinal with m modalities), it provides $(m-1)$ possible binary divisions.
- If it qualitative but not ordinal, the number of divisions becomes $2^{(m-1)}-1$.
- The division criterion is based on the definition of an heterogeneity function.
- The objective is to divide the observations which compose a node into two more homogeneous groups with respect to the variable to explain Y.

Division criterion

Optimal division

- Dividing the node κ creates two son nodes : κ_L (left node) and κ_R (right node).
- Given an heterogeneity function D_κ of the node κ, the algorithm retains the division which minimizes

$$D_{\kappa_L} + D_{\kappa_R}.$$

- For each node κ in the construction of the tree :

$$\max_{\{Divisions\ of\ X^j; j=1,p\}} D_\kappa - (D_{\kappa_L} + D_{\kappa_R})$$

- Graphically, the length of each branch can be represented proportionally to the reduction in heterogeneity induced by the division.

Stopping rule and affectation

- The growth of the tree stops at a given node, which therefore becomes a terminal node also called a *leaf*,
  - when it is homogeneous (all the individuals have the same value for Y) or
  - when there is no longer an admissible partition or
  - to avoid unnecessarily fine splittings, when the number of observations it contains is less than some prescribed value (generally chosen between 1 and 5).
- When Y is quantitative, the predicted value associated to a leaf is the mean of the values of the Y_i's among the observations belonging to this terminal node.
- In the qualitative case, each leaf or terminal node is assigned to a class $\mathcal{T}_\ell$ of Y by a majority vote.

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Constructing regression trees

For a given region (node) κ with size $|\kappa|$, define the heterogeneity by :

$$D_\kappa = \sum_{i \in \kappa} (y_i - \overline{y}_\kappa)^2 = |\kappa| \frac{1}{|\kappa|} \sum_{i \in \kappa} (y_i - \overline{y}_\kappa)^2$$

Splitting procedure

For a variable x_j, and a split candidate t, define left and right subregions

$$\kappa_L(t,j) = \{x_j \leq t\}, \qquad \kappa_R(t,j) = \{x_j > t\}.$$

Find (j,t) in order to minimize the intra-class variance

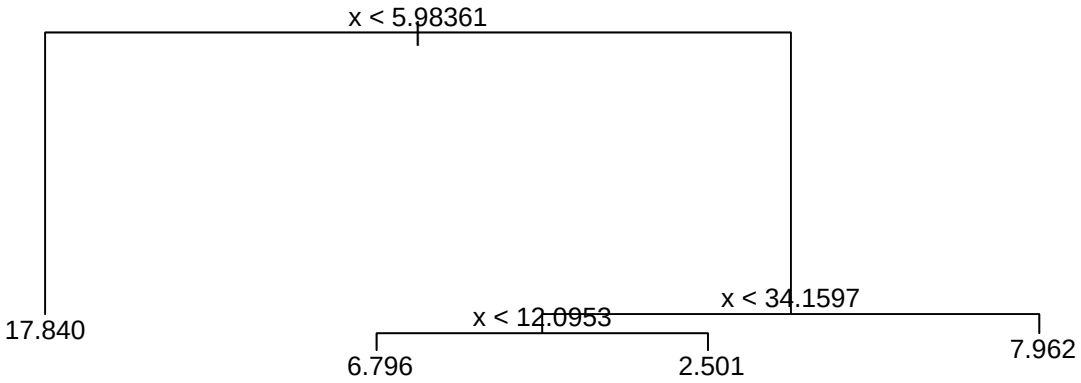
$$J(j,t) = D_{\kappa_L(t,j)} + D_{\kappa_R(t,j)},$$

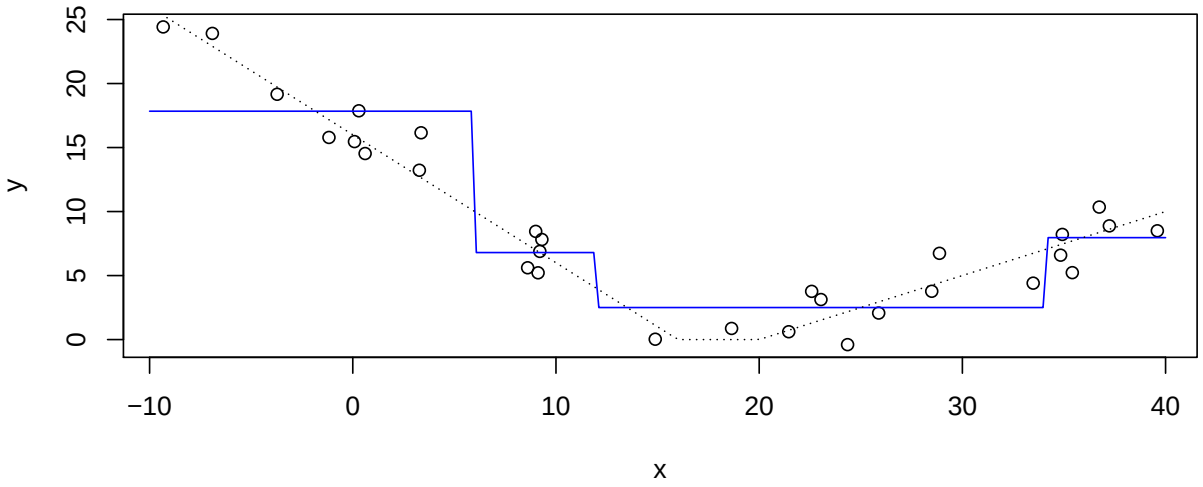
or equiv. to maximize the decrease in heterogeneity (inter-class variance)

$$D_\kappa - J(j,t)$$

NB : To remove ties, the best cut t is taken at the middle of two consecutive data points.

Illustration in 1 dimension





Constructing classification trees

This is the same procedure, with specific notions of heterogeneity

Heterogeneity measures in classification

p_κ^ℓ : proportion of the class $\mathcal{T}_\ell$ of Y in the node κ.

- The Cross-Entropy or deviance is defined by

$$E_\kappa = -\sum_{\ell=1}^{m} p_\kappa^\ell \log(p_\kappa^\ell) \qquad \Rightarrow \qquad D_\kappa = -|\kappa| \sum_{\ell=1}^{m} p_\kappa^\ell \log(p_\kappa^\ell)$$

  Maximal in $(\frac{1}{m}, \ldots, \frac{1}{m})$, minimal in $(1, 0, \ldots, 0), \ldots, (0, \ldots, 0, 1)$
  (by continuity, we assume that $0 \log(0) = 0$)

- Gini concentration : $D_\kappa = |\kappa| \sum_{\ell=1}^{m} p_\kappa^\ell (1 - p_\kappa^\ell)$

Illustration with two classes ($m = 2$)

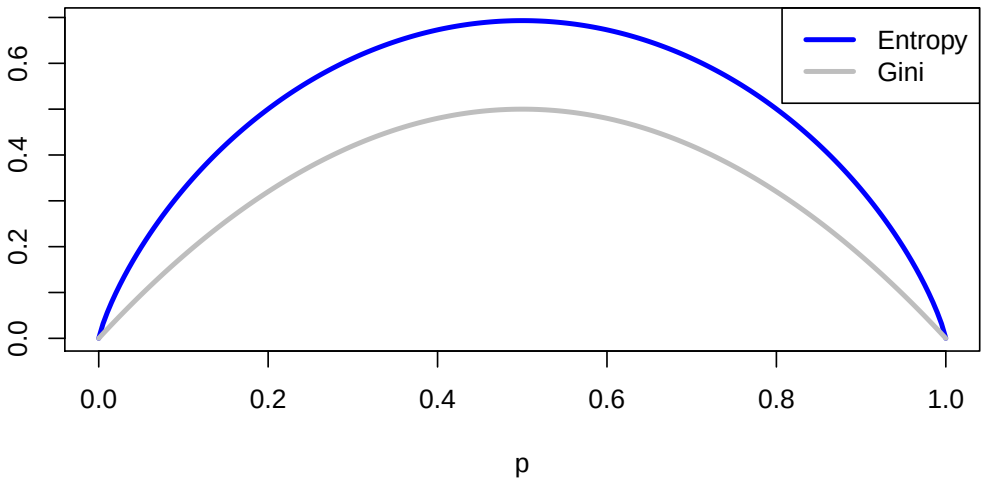


Figure – Heterogeneity criterions for classification. Both are minimal for $p = 0$ or $p = 1$, and maximal for $p = 1/2$.

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Stopping rule, pruning, optimal tree

- We need a tradeoff between maximal tree (overfits) and the constant tree (too rough)
- There exists a nice theory to find an optimal tree, minimizing prediction error penalized by complexity (number of leaves)
- When aggregating trees (random forest), simpler procedures are often preferred (see why after), e.g. fixing the number of leaves.

Pruning and optimal tree

Pruning : notations

- We look for a parcimonious tree
- *Complexity* of a tree : $|A|$ = numbers of leaves in A
- Adjustment error of A :

$$D(A) = \sum_{\kappa \text{ leaf of } A} D_\kappa$$

D_κ : heterogeneity of leaf κ

Sequence of embedded trees

- Adjustment error penalized by the complexity :

$$Crit_\gamma(A) = D(A) + \gamma \times |A|$$

- When $\gamma = 0$: $A_{\max}$ (maximal tree) minimizes $Crit_\gamma(A)$
- When γ increases, the division for which the improvement of D is smaller than γ, is cancelled ; hence
  - two leaves are gathered (prunned)
  - there father node becomes a terminal node
  - A_J becomes A_{J+1}.
- After iteration of this process, we get a sequence of trees :

$$A_{\max} \supset A_1 \supset A_2 \supset \cdots A_K$$

Breiman sub-sequence

- A_1 is the sub tree of $A_{\max}$ (maximal tree) obtained by pruning the nodes κ such that $D(\kappa) = D(\kappa_L) + D(\kappa_R)$.
- For each node κ in A_1, $D(\kappa) > D(\kappa_L) + D(\kappa_R)$ and by induction, $D(\kappa) > D(A_1^\kappa)$ where A_1^κ is the subtree of A_1 from node κ.
- For γ small, $D(\kappa) + \gamma > D(A_1^\kappa) + \gamma|A_1^\kappa|$. This holds while $\gamma < (D(\kappa) - D(A_1^\kappa))/(|A_1^\kappa| - 1) = s(\kappa, A_1^\kappa)$ for all node κ of A_1.

$$\gamma_1 = \inf_{\kappa \text{ node of } A_1} s(\kappa, A_1^\kappa) = s(\kappa^*, A_1^{\kappa^*})$$

- $Crit_{\gamma_1}(\kappa^*) = Crit_{\gamma_1}(A_1^{\kappa^*})$ and the node κ^* becomes preferable to the subtree $A_1^{\kappa^*}$.
- A_2 is the subtree obtained by pruning the branches from the nodes minimizing $s(\kappa, A_1^\kappa)$: this gives the second tree in the sub-sequence
- We iterate this process.

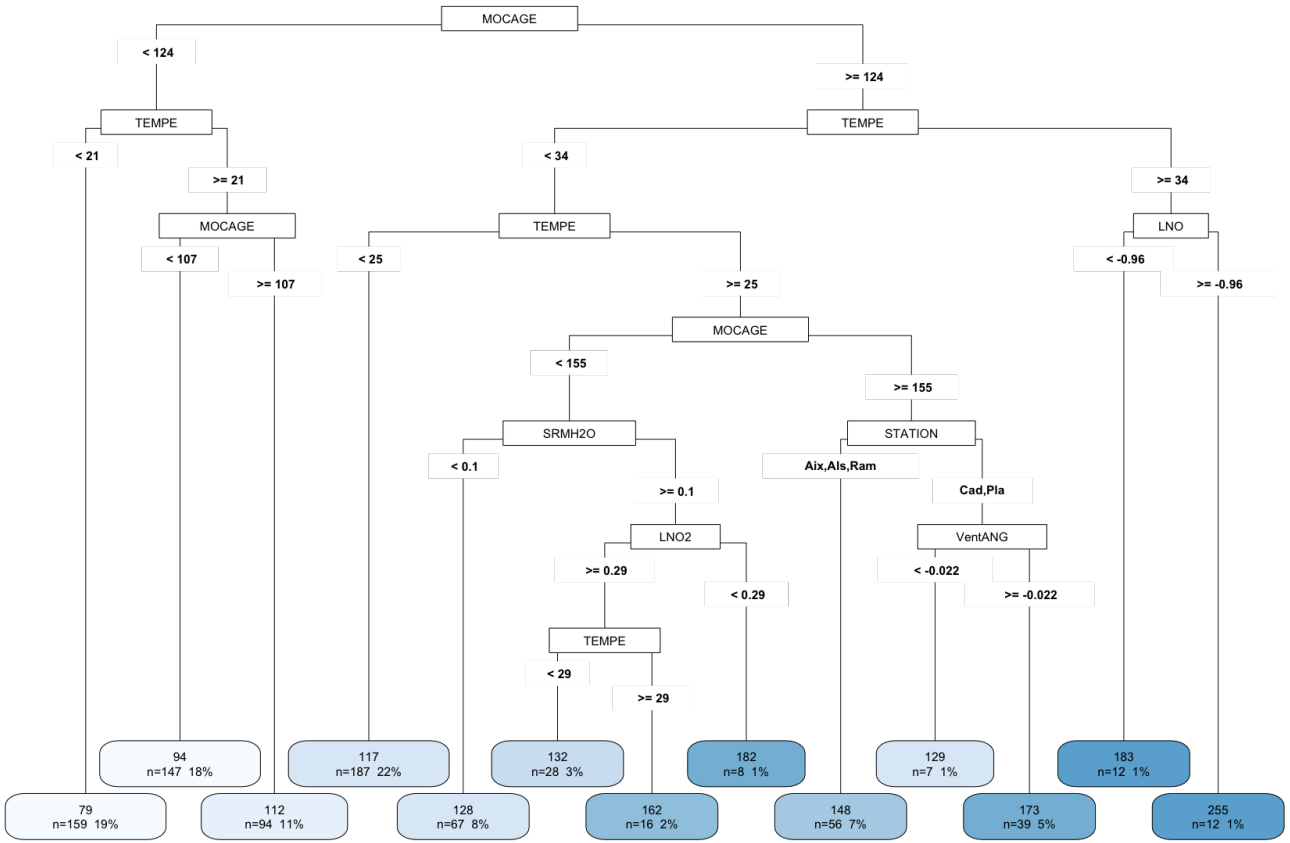
Optimal tree

Algorithm to select the optimal tree

- Maximal tree $A_{\max}$
- Construction of Breiman's sequence $A_1 \dots A_K$ of nested trees associated with the sequence of penalization parameters $(\gamma_1, \dots, \gamma_K)$
- V-fold cross validation error :
  for $v = 1, \dots, V$ **do**
  - For each sample (composed of $V - 1$ folds), estimation of the sequence of trees associated with the sequence of penalization parameters $(\gamma_1, \dots, \gamma_K)$
  - Estimation of the error on the validation fold.

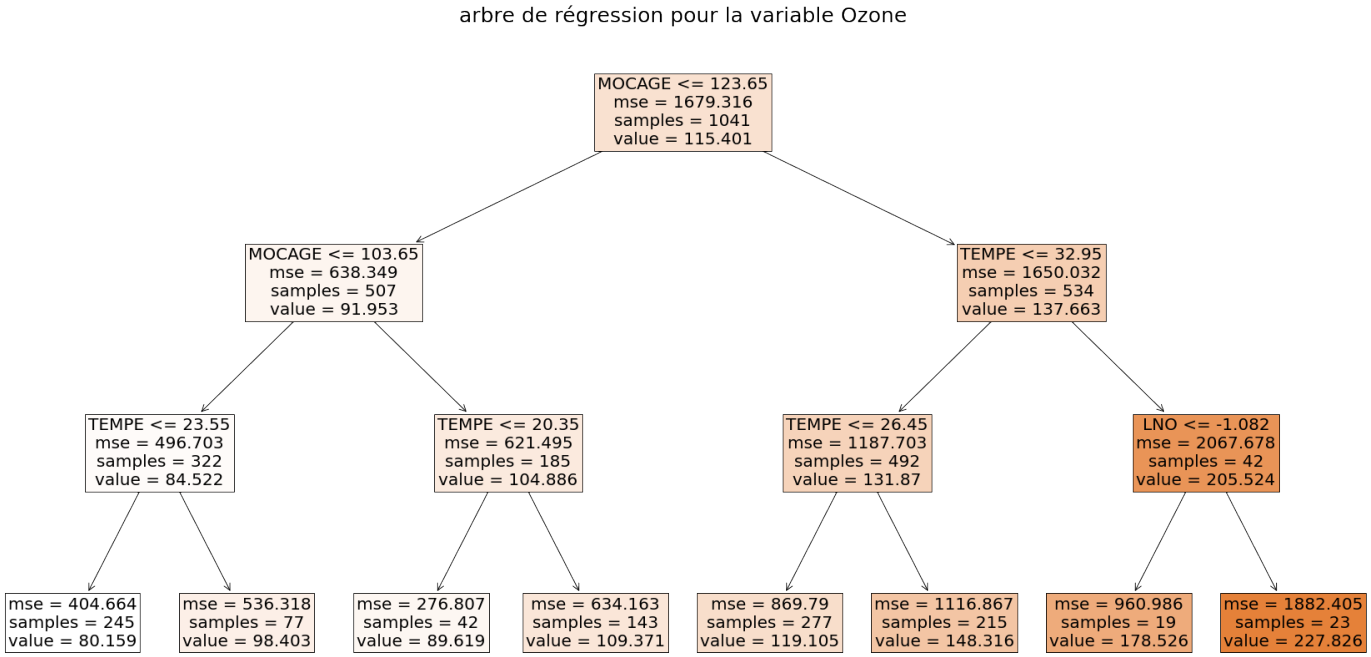
  EndFor
- For each value $(\gamma_1, \dots, \gamma_K)$, computation of the means of these errors.
- Determination of the optimal value γ_{Opt}, corresponding to the minimal error mean.
- Retain the tree corresponding to γ_{Opt} in Breiman's subsequence $A_1 \dots A_K$

Example of pruned tree for Ozone data



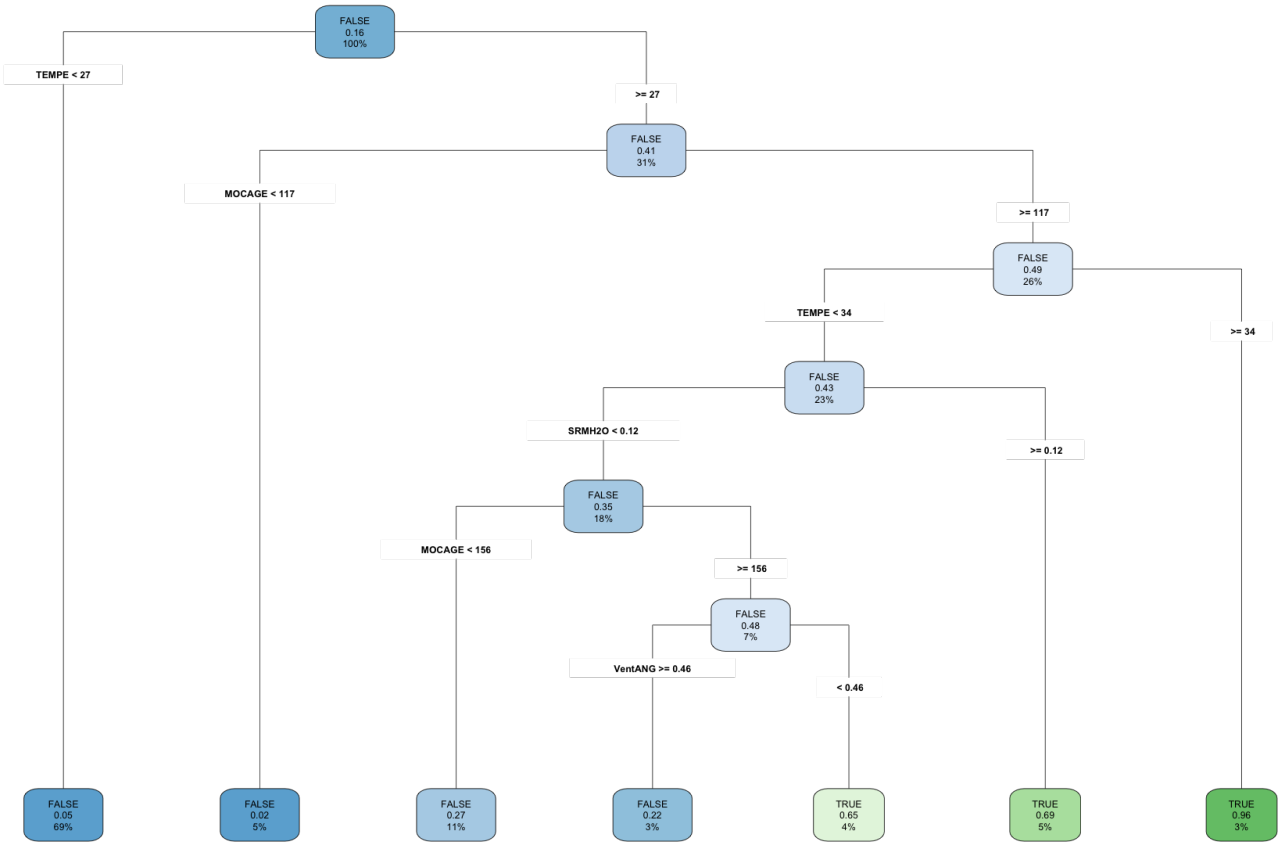
Ozone : Regression tree pruned by cross-validation(R software)

Example of pruned tree for Ozone data



Ozone : Regression tree pruned by cross-validation (Python-sklearn)

Example for Ozone data



Ozone : Classification tree pruned by cross-validation

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Missing values

- CART is tolerant to missing data for the prediction phase.
- Assume that the dataset has some missing predictor values for some (or all) of the variables.
- Instead of discarding observations with missing values, or imputing the missing values, CART proposes two better strategies.
  - First, for categorical variables, we can add a category for "missing".
  - The second approach is to construct surrogate variables that will be considered if the value of a variable is missing. We choose as usual the best predictor and split at one node, the first surrogate is the second best, and so on.. When an observation is sent down the tree (during the training or prediction phase), if the value of a predictor is missing at one node, we use the first surrogate ; if this one is missing, we use the second and so on..

Missing values

- CART is also tolerant to missing data in the learning phase :
- The data with missing value for Y or for all the explanatory variables are eliminated.
- The decrease in heterogeneity for each variable and split is computed by using the associated available observations and the best split is chosen as usual.

Instability of trees

- A major drawback of trees is their high variance. They are not robust, in the sense that a small change in the data can lead to very different sequences of splits.
- This is why we have to be careful with the interpretation. This is due to the hierarchical procedure : an error in the choice of a split in the top of the tree cannot be corrected below.
- This instability is the price to pay to have a simple and interpretable model.
- We will see in the next chapter how to aggregate trees to obtain Random Forests and reduce the variance of the prediction rule.

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Advantages

- Trees are easy to interpret
- Efficient algorithms to find the pruned trees
- Tolerant to missing data

$\Longrightarrow$ Success of CART for practical applications

Warnings

- Variable selection : the selected tree only depends on few explanatory variables, trees are often (wrongly) interpreted as a variable selection procedure
- High instability of the trees : not robust to the learning sample, curse of dimensionality ..
- Prediction accuracy of a tree is often poor compared to other procedures

$\Longrightarrow$ This is why Random Forests have been introduced.

References

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